MAYANK RAJ

Ranchi, India | +91 92340 56069 | mayank1312.raj@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Science undergraduate at BIT Mesra (graduating May 2027) and GATE 2026 qualifier with hands-on experience in backend integration, full-stack development, and geospatial AI. Delivered a Sentinel-1 SAR flood-mapping pipeline, an NLP resume-matching tool, and REST APIs using Python, FastAPI, Node.js, React, and MongoDB.

EDUCATION

Birla Institute of Technology, Mesra, Ranchi — Aug 2023 – May 2027
B.Tech. in Computer Science and Engineering — *CGPA: 8.0/10*

Jawahar Vidya Mandir, Shyamali, Ranchi — Apr 2020 – Apr 2022
CBSE Class XII – Science

EXPERIENCE

Birla Institute of Technology, Mesra — May 2026 – Jul 2026
Undergraduate Researcher – Geospatial AI and Disaster Analysis — *Ranchi, India*

- Engineered a reproducible **Sentinel-1 GRD VV SAR flood-mapping pipeline** covering **14 Kerala districts**, estimating **344.74, 273.28, and 219.42 km^2** of inundation at -2.0, -2.5, and -3.0 dB thresholds.
- Reduced mapping artifacts with JRC permanent-water and SRTM slope masks, enabling direct comparison of sensitive, balanced, and conservative flood estimates.
- Designed an **NLP and SAR verification workflow** over **7,984 English, Malayalam, and Romanized posts**, extracting locations, sentiment, urgency, and claim types to measure agreement between social-media flood reports and satellite evidence.
- Co-Authored a **research paper selected for presentation at IEEE DELCON 2026**, based on the flood-mapping and social-media verification study.

Mecon Ltd. — Dec 2025 – Jan 2026
Frontend Web Development Intern — *Ranchi, India*

- Built a **React pathfinding visualizer** on a custom hexagonal grid, implementing **four graph algorithms** (A*, BFS, DFS, Dijkstra's) with interactive controls that let users compare how each explores the grid.
- Organized the codebase into **reusable React components** and component-scoped **SCSS**, separating grid rendering, controls, and layout for easier maintenance and extension.

Ransh Innovations Pvt. Ltd. GitHub — Dec 2024 – Aug 2025
Full Stack Web Development Intern – Backend Focus — *Remote*

- Unified tournament information across **10+ sports** from **3+ external REST API providers**, transforming provider-specific responses into a consistent format for the company's website and enabling tournament discovery in one place.
- Implemented **daily API request controls** with a **100-request limit** and reused stored tournament data for repeat visits, avoiding duplicate external fetches while keeping requests within the configured quota.

PROJECTS

MINISTA – Full-Stack Social Media Platform — Live Demo | GitHub

- Developed a **MERN-stack social platform** using React, Node.js, Express.js, and MongoDB, implementing **6 core workflows**: authentication, post creation, ImageKit uploads, feeds, likes, and follow management.
- Implemented **2 feed modes**—global and personalized—with REST APIs; secured protected routes using cookie-based JWT authentication and configured CORS for separate frontend and backend deployments.

AI-Powered CV Checker – Resume and Job Matching — Live Demo | GitHub

- Built a **React and FastAPI resume analyzer** supporting **2 document formats** (PDF and DOCX) and extracting **4 information categories**: skills, experience, education, and contact details.
- Combined **NLP-based semantic similarity** with job-description keyword analysis to identify missing skills and generate targeted resume-improvement feedback.

SKILLS

Languages: Java, Python, C, JavaScript, TypeScript, SQL
Frontend: React.js, Vite, Tailwind CSS, SCSS
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, JWT
Databases: MongoDB, MySQL, ChromaDB
AI/ML: Machine Learning, NLP, Semantic Similarity, RAG, LangChain, TensorFlow, Scikit-learn
Tools: Git, GitHub, Docker, Postman, Vercel, Render
Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Communication: Technical documentation, research writing

CERTIFICATIONS

Machine Learning with AI Training — 2024
Internshala Trainings

IBM Generative AI, RAG Applications and Vector Databases for RAG — 2026
Coursera